

FREISTADT, Austria

WMO: 110150

Lat: 48.507N

Lon: 14.507E

Elev: 548

StdP: 94.91

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-12.0	-16.8	0.9	-14.8	-13.8	1.2	-11.6	5.5	1.7	4.6	2.1	0.8	10	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	29.6	20.2	27.6	19.1	25.8	18.1	20.8	28.0	19.8	26.2	18.8	24.3	1.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	14.1	23.7	17.4	13.3	22.4	16.7	12.7	21.5	62.5	28.1	58.8	26.4	55.5	24.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.4	3.8	3.2	-18.2	32.7	3.3	1.5	-20.6	33.8	-22.5	34.7	-24.3	35.5	-26.7	36.6
			-18.4	22.5	3.1	1.1	-20.6	23.3	-22.5	23.9	-24.2	24.5	-26.5	25.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-2.8	-0.6	3.1	8.3	12.8	16.0	17.7	17.5	12.8	8.1	3.0	-1.1
	DBStd	8.16	4.62	4.53	4.16	4.18	3.62	3.74	3.21	3.14	3.29	3.80	3.81	4.11
	HDD10.0	1655	398	297	214	79	14	2	0	1	11	82	212	344
	HDD18.3	3878	656	531	472	302	174	90	51	53	168	317	461	603
	CDD10.0	900	0	0	1	27	101	181	239	232	94	24	1	0
	CDD18.3	81	0	0	0	0	2	19	32	26	2	0	0	0
	CDH23.3	1028	0	0	0	8	62	253	365	309	30	1	0	0
	CDH26.7	256	0	0	0	1	6	67	103	77	2	0	0	0
Wind	WSAvg	1.2	1.2	1.3	1.5	1.5	1.4	1.3	1.2	1.2	1.1	1.1	1.0	1.0
Precipitation	PrecAvg	803	42	40	56	51	82	101	113	105	66	49	48	47
	PrecMax	1201	82	96	121	93	177	198	207	398	174	117	87	84
	PrecMin	637	7	9	15	9	16	43	57	47	15	11	6	11
	PrecStd	116	23	24	28	26	33	38	36	68	34	30	20	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	12.5	18.0	24.2	27.4	31.0	32.1	31.8	26.2	21.6	16.2	9.7
		MCWB	6.8	7.6	10.2	14.5	17.8	20.7	21.4	20.9	18.1	15.3	11.5	7.3
	2%	DB	7.4	9.5	15.0	21.3	25.2	28.9	29.4	28.9	23.9	18.6	12.5	7.5
		MCWB	5.4	6.2	9.3	13.3	16.7	19.8	20.1	20.1	17.2	13.7	9.3	5.9
	5%	DB	5.4	7.5	12.4	18.8	23.2	26.4	27.4	26.6	21.7	16.4	10.2	5.6
		MCWB	4.0	4.8	7.9	11.9	15.6	18.5	19.2	18.7	16.0	12.6	8.3	4.3
	10%	DB	3.6	5.6	9.8	16.5	21.0	24.0	25.1	24.8	19.5	14.5	8.4	4.0
		MCWB	2.4	3.6	6.7	10.7	14.6	17.2	18.2	17.8	15.0	11.6	6.9	3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.3	11.0	15.2	18.6	21.7	22.1	22.3	19.0	15.7	11.6	7.6
		MCDB	9.2	11.5	16.3	23.2	25.9	29.2	30.3	30.1	24.8	20.2	15.5	8.8
	2%	WB	5.6	6.4	9.7	13.6	17.3	20.3	20.7	20.5	17.5	14.2	9.7	5.9
		MCDB	7.3	8.8	14.2	20.3	24.0	27.5	27.9	27.5	22.6	17.8	11.8	7.3
	5%	WB	4.1	5.1	8.4	12.3	16.1	19.1	19.8	19.3	16.4	13.1	8.4	4.5
		MCDB	5.4	7.2	11.8	17.9	22.1	25.2	26.0	25.3	20.7	15.8	10.1	5.6
	10%	WB	2.5	3.7	7.0	11.2	15.0	17.9	18.7	18.3	15.4	11.9	7.0	3.0
		MCDB	3.5	5.5	9.6	15.9	20.1	22.8	23.9	23.5	18.9	14.1	8.3	3.9
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.9	9.2	10.9	11.1	10.9	10.9	11.1	10.2	9.8	6.9	5.9
		MCDBR	8.0	11.1	14.7	16.0	16.1	15.8	15.7	15.4	14.3	13.1	10.8	8.2
		MCWBR	6.3	7.8	9.0	8.5	8.1	7.2	7.0	7.1	7.8	8.4	7.6	6.5
	5% WB	MCDBR	7.3	10.1	13.6	14.7	14.8	14.3	13.9	14.2	12.9	11.9	10.1	7.5
		MCWBR	5.9	7.2	8.6	8.1	7.6	6.8	6.4	6.9	7.3	7.9	7.3	6.1
Clear-Sky Solar Irradiance	taub	0.289	0.314	0.353	0.390	0.395	0.398	0.406	0.401	0.374	0.344	0.318	0.290	
	taud	2.390	2.356	2.319	2.275	2.297	2.318	2.295	2.326	2.373	2.440	2.465	2.447	
	Ebn at Noon	781	839	857	858	867	864	850	838	826	794	741	736	
	Edh at Noon	67	87	106	123	125	124	125	116	100	80	63	56	
All-Sky Solar Radiation	RadAvg	0.99	1.88	2.94	4.31	4.99	5.45	5.26	4.69	3.34	2.08	1.05	0.78	
	RadStd	0.16	0.25	0.38	0.53	0.58	0.48	0.51	0.47	0.43	0.24	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (96 neighbors)	+0.50	N/A	N/A	N/A	+0.39	+0.34	N/A	N/A	+68	+30

Nomenclature: See separate page